

Detecting Rail Defects with AI

A hands-on introduction to image segmentation and model tuning

No programming experience required

You will follow copy-paste steps in a terminal and edit one plain-text settings file. That's it – the AI code itself is already written for you.

What you'll learn

- What “Non-Destructive Testing” (NDT) means and why AI can help with it
- What it means for AI to “segment” an image (draw where something is, pixel by pixel) instead of just labeling the whole photo
- How a single setting – the **decision threshold** – controls the trade-off between catching every defect and avoiding false alarms
- How to read an accuracy score (IoU) and compare it with classmates

The big picture

Railway maintenance crews inspect rail surfaces for defects (cracks, spalling, corrosion) that could cause problems if left unnoticed – traditionally by a human walking the track and looking. NDT means checking for defects without damaging the rail (e.g. just photographing it), which is exactly what our dataset does.

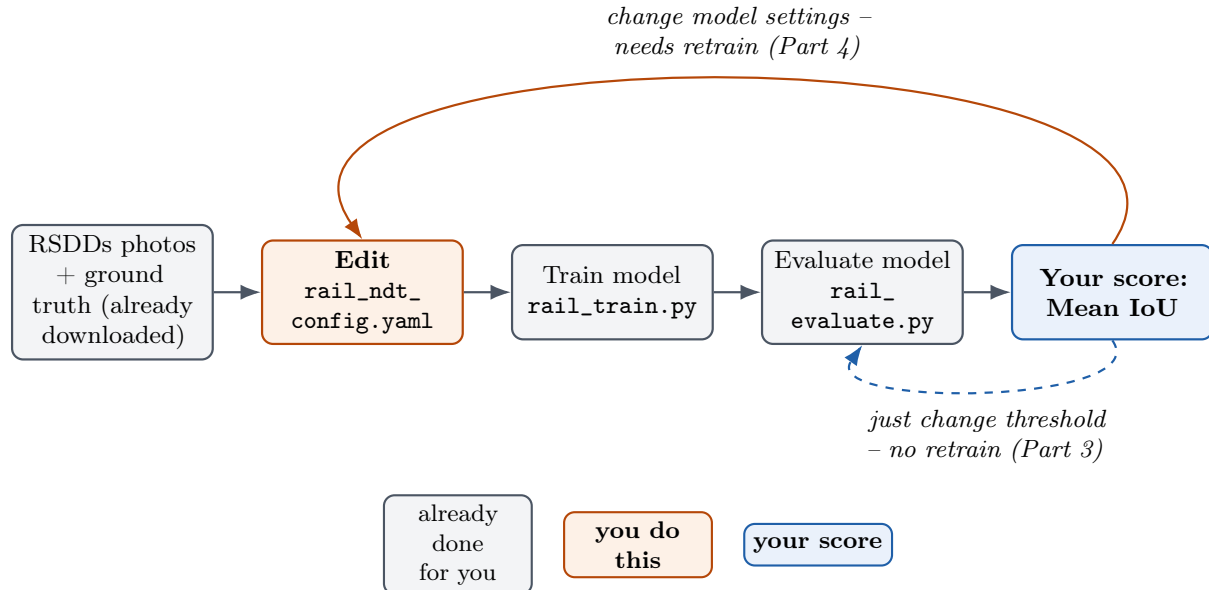
Key idea

Unlike Lab 1, **nobody needs to label anything here** – the dataset already comes with 113 real rail photos, each with a hand-drawn “ground truth” map showing exactly where the defect is (drawn by the researchers who published this dataset). Your job is to train an AI that draws its own version of that map on a new photo, and to make that AI as accurate as possible.

What's already done for you

- The dataset (113 photos + their official defect maps) has already been downloaded into `datasets/raw/rail_ndt/rsdds/`.
- All the AI code – feature extraction, model training, scoring, plotting – is already written in `scripts/`. You will only *run* it, never edit it.
- The only file you'll edit is `configs/rail_ndt_config.yaml`, a plain-text settings file (not code).

How it all fits together



Two different loops back into the same pipeline: the dashed blue one under the diagram is fast (just re-run evaluate); the solid orange one over the top requires starting again from training.

Part 0: One-time setup

Open a terminal (on Mac: the **Terminal** app; on Windows: **Anaconda Prompt** or **PowerShell**) and run these commands one at a time:

```
cd AI-training
python3 -m venv .venv
source .venv/bin/activate # Windows: .venv\Scripts\activate
pip install -r requirements.txt
```

If you already did this for Lab 1 in the same folder, you can skip straight to Part 1 (just remember to run the `source .venv/bin/activate` line again in any new terminal window).

Part 1: Train your model

Open `configs/rail_ndt_config.yaml` in any plain-text editor (Notepad, TextEdit, VS Code – anything that isn't Word). You'll see settings like:

```
model_type: random_forest
n_estimators: 100
max_depth: 8
random_seed: 42
threshold: 0.5
```

For now, leave everything as-is and run:

```
python3 scripts/rail_train.py
```

This looks at every pixel of the training photos and learns what a “defect pixel” tends to look like (its brightness, texture, and where in the photo it is), compared to normal rail surface. It takes under a minute.

Part 2: Evaluate your model

```
python3 scripts/rail_evaluate.py
```

This tests your model on photos it never trained on, and prints something like:

```
Mean IoU: 0.25 (higher is better, 1.0 = perfect)
Mean Dice: 0.38 (higher is better, 1.0 = perfect)
```

IoU (Intersection over Union) measures how much your model’s guessed defect area overlaps with the real defect area. 1.0 = perfect overlap, 0 = no overlap at all. This is your score – **higher is better**.

It also saves two pictures:

- `results/figures/rail_eval.png` – several test photos side by side, each with the true and predicted defect areas overlaid directly on the rail photo in color – **yellow** = correctly caught, **green** = a real defect the model missed, **red** = a false alarm – with that photo’s own IoU score labeled above it. Look at it – this is the most useful way to understand what your model is actually doing.
- `results/figures/rail_eval_threshold_curve.png` – Mean IoU and Mean Dice swept automatically across every threshold from 0.05 to 0.95 for this same trained model, with your current threshold marked. It’s a preview of the trade-off you’ll explore by hand in Part 3.

Part 3: Tune the threshold (the main lever – no retraining needed)

Your model doesn’t just say “defect” or “not defect” for each pixel – it gives a confidence score from 0 to 1. The `threshold` setting decides how confident it must be before marking a pixel as a defect:

threshold	Effect
Low (e.g. 0.2)	Flags more pixels – catches more real defects, but also more false alarms
High (e.g. 0.8)	Flags fewer pixels – fewer false alarms, but risks missing real defects

Open `configs/rail_ndt_config.yaml`, change only the `threshold` value, and re-run **just** the evaluate step (no need to retrain):

```
python3 scripts/rail_evaluate.py
```

Tip

Try several values (e.g. 0.3, 0.4, 0.5, 0.6, 0.7, 0.8) and note the Mean IoU each time. This is the fastest way to improve your score, and needs no retraining – each check takes only a few

seconds.

Part 4: (Optional/stretch) Retrain with different settings

If you want to go further, edit `model_type`, `n_estimators`, `max_depth`, or `random_seed`, then re-run **both** steps in order (these settings only take effect after retraining):

```
python3 scripts/rail_train.py
python3 scripts/rail_evaluate.py
```

Then repeat Part 3 to re-tune the threshold for your new model – the best threshold can be different for different settings.

Compete with classmates

Keep a scoreboard:

Your name	model_type	key settings	threshold	Mean IoU (higher=better)
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Part 5 (optional): Generate a shareable report

Once you're happy with a run, turn it into a one-page PDF you can print or send to a classmate/instructor:

```
python3 scripts/rail_report.py
```

This reads your latest results and saves `results/reports/rail_report.pdf` – your settings, score, example predictions, and the threshold curve on a single page. Re-run it any time after Part 2 (or Part 3's threshold tweaks) to capture your current best result.

Glossary

- **NDT (Non-Destructive Testing)** – inspecting something for defects without damaging it, e.g. from a photo instead of cutting the rail open.
- **Segmentation** – instead of labeling a whole photo (“defective” or not), the model marks *which pixels* belong to the defect.
- **Ground truth** – the correct answer, here a hand-drawn map of exactly where each defect is, made by the dataset’s original researchers.
- **Threshold** – the confidence cutoff (0 to 1) above which a pixel is called “defect.” The only setting you can change without retraining.

- **IoU (Intersection over Union)** – overlap between your model’s guessed defect area and the real one, divided by their combined area. 1.0 = identical, 0 = no overlap.
- **Dice score** – a very similar overlap measure to IoU; usually a bit higher for the same prediction. Both are reported so you can compare.
- **False positive** – a normal pixel wrongly marked as a defect (a false alarm).
- **False negative** – a real defect pixel the model missed.
- **Class imbalance** – in every photo, the vast majority of pixels are normal rail and only a small area is defect. The model is specifically built to account for this, otherwise it could get a deceptively high score just by never predicting “defect” at all.
- **Random Forest** – a model made of many simple decision-making “trees” that vote on each pixel; `n_estimators` = number of trees, `max_depth` = how detailed each tree’s questions can get.

Troubleshooting

- **“no RSDDs images found”** – run `python3 scripts/fetch_rsdds.py` first to download the dataset (only needs to be done once).
- **“no trained model found”** – run `python3 scripts/rail_train.py` before `rail_evaluate.py`.
- **Changed a setting but the score didn’t change** – only `threshold` takes effect immediately; every other setting requires re-running `rail_train.py` first.
- **Score looks much worse than a classmate’s with identical settings** – double check `random_seed` is the same; it controls which photos land in the training vs. test group, so a different seed is training/testing on a different split of photos entirely.

Something to think about (discuss with the class)

- Why is a high IoU score harder to achieve here than a high accuracy score would be, given how small the defect areas are compared to the whole photo?
- In a real inspection system, would you rather have a lower threshold (more false alarms) or a higher one (more missed defects)? Does your answer change if a missed crack could cause a derailment?
- This dataset has no “clean rail” photos without any defect at all – every image contains one. How might that affect what the model has learned, and what would you want to add to fix it?